BA-BE Practice 1

Module 7a - Bioavailability and Bioequivalence · Daily practice sheet · No date

Name _____ Date _____

The equation for concentration of drug in the plasma as a function of time following a 250 mg IV bolus dose was found to be: $C_p = 30e^{-0.092t}$.

a If the AUC following a 500 mg oral tablet dose was found to be 435 mcg·hr/mL, what is the bioavailability of the drug in the tablet dosage form?

b What oral dose would provide comparable bioavailability to the 250 mg IV bolus dose? (Round to the nearest 5 mg) (The tablet is 0.667 bioavailable.)

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 0.667 (as a decimal fraction)

WHAT THE STEM GIVES

D IV	250 mg	$V_D = \frac{250}{30} = 8.33$ L, and the dose ratio $\frac{250}{500}$
C_0	30	the IV intercept, mg/L: $V_D = \frac{250}{30} = 8.33$ L
k	0.092	the IV slope, hr^{-1}: $Cl_T = k \times V_D = 0.766$ L/hr
D po	500 mg oral tablet	the denominator of F
AUC po	435 mcg·hr/mL	435 (mg/L)hr, multiplied by Cl_T

THE WORKING, ONE LINE AT A TIME

UNIT

$$435 \text{ mcg·hr/mL} = 435 \text{ (mg/L)hr};\ V_D = \frac{250 \text{ mg}}{30 \text{ mg/L}} = 8.33 \text{ L}$$

1 mcg/mL is 1 mg/L, so the oral area is 435 (mg/L)hr; the volume is the IV dose over the intercept of the IV curve.

SETUP

$$F \text{ D po} = Cl_T \times \text{AUC po, so } F = \frac{\text{AUC po} \times k \times V_D}{\text{D po}}$$

Clearance times the area under the oral curve is the amount that reached the circulation, F × D po; with $Cl_T = k\ V_D$, F follows from the IV parameters and the oral area.

ALGEBRA

$$F = \frac{(435 \text{ (mg/L)hr})(0.092\ hr^{-1})(8.33 \text{ L})}{500 \text{ mg}} = \frac{333.4 \text{ mg}}{500 \text{ mg}} = 0.667$$

(mg/L)hr × hr^{-1} × L leaves mg, and mg over mg is a pure fraction: two thirds of the tablet dose reached the circulation.

ALGEBRA

$$\text{Or } F = \frac{\text{AUC po}}{\text{AUC IV}} \times \frac{\text{D IV}}{\text{D po}} = \frac{435 \text{ (mg/L)hr}}{326 \text{ (mg/L)hr}} \times \frac{250 \text{ mg}}{500 \text{ mg}} = 0.667$$

The two-AUC form gives the same value, with AUC IV = 30 over 0.092 = 326 (mg/L)hr and the dose ratio correcting for the different doses.

> **DOES THE NUMBER MAKE SENSE?**
>
> F lies between 0 and 1. The IV area is 30 over 0.092, about 326 (mg/L)hr, from 250 mg; the oral 500 mg gave 435, less than twice 326, so F is below 1 and near two thirds: 0.667.

b

Her key

Answer: 375 mg

WHAT THE STEM GIVES

IV curve	Cp = 30e^(−0.092t)	*not needed: F is already given*
D IV	250 mg	the numerator of D po = $\frac{\text{D IV}}{\text{F}}$
rounding	nearest 5 mg	rounds 374.8 to 375 mg
F	0.667	the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{F} \times \text{D po} = \text{D IV}, \text{ so D po} = \frac{\text{D IV}}{\text{F}}$$

Comparable bioavailability means the oral AUC equals the IV AUC, so the fraction absorbed times the oral dose must equal the whole 250 mg IV dose.

ALGEBRA

$$\text{D po} = \frac{250 \text{ mg}}{0.667} = 374.8 \text{ mg}$$

F has no units, so the oral dose is in milligrams; it is larger than 250 mg because only two thirds of the tablet dose reaches the circulation.

ROUND

$$\text{D po} = 375 \text{ mg}$$

The stem asks for the nearest 5 mg, so 374.8 mg is reported as 375 mg, a quantity that can be made up from tablet strengths. Check: 0.667 × 375 mg = 250.1 mg reaches the circulation, matching the 250 mg IV dose, so the two routes give the same AUC.

> **DOES THE NUMBER MAKE SENSE?**
>
> Two thirds of the oral dose reaches the circulation, so the oral dose is half as much again as 250 mg: 375 mg. The oral dose must exceed 250 mg because F is below 1.